

new title

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ABSTRACT

abstract

Keywords: word1 — word2 — word3 — word4 — word5

INTRODUCTION

The PageRank algorithm, first introduced by Larry Page and Sergey Brin, was one of the first algorithms to rank webpages based on their link structure rather than just content [Brin & Page \(1998\)](#). The algorithm models the web as a directed graph and simulates a "random surfer" who follows hyperlinks with some probability or jumps to a random page to avoid getting stuck [Page et al. \(1999\)](#).

This setup defines a Markov process where each page's importance is proportional to the importance of pages linking to it. The final PageRank vector corresponds to the steady-state distribution of this process and can be found either by iteration or by solving for the dominant eigenvector of the hyperlink matrix $\mathbf{H}$ [Langville & Meyer \(2009\)](#).

In this project, we first implemented PageRank using the difference equation $x_{k+1} = \mathbf{H}x_k$, which resulted in the scores converging once the solution reached the maximum tolerance allowed. We then confirmed our result by solving for the eigenvector associated with the largest eigenvalue of the matrix $\mathbf{H}$. Our goal was to understand how the algorithm behaves under different network structures and how quickly it converges [Langville & Meyer \(2004\)](#).

DATA AND OBSERVATIONS

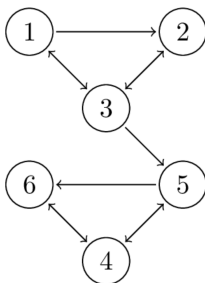


Figure 1. example graph 1.

To start, we defined a 6-node directed graph where each page links to one or more pages, as shown in Figure 1. From this, we constructed the hyperlink matrix $\mathbf{H}$, where each non-zero entry $\mathbf{H}_{ij}$ is equal to $\frac{1}{|\mathcal{P}_{ij}|}$, the reciprocal of the number of outbound links from page P_j . This matrix represents the probability that a random surfer on page j clicks through to page i :

$$\mathbf{H} = \begin{pmatrix} 0 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/3 & 0 & 0 & 0 \\ 1/2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1/2 & 1 \\ 0 & 0 & 1/3 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \quad (1)$$

We initialized the PageRank vector $\mathbf{x}_0$ with equal values and assumed that each page is equally likely to be visited at first:

$$\mathbf{x}_0 = \begin{pmatrix} 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \\ 1/6 \end{pmatrix} \quad (2)$$

We then applied the iterative update rule $\mathbf{x}_{n+1} = \mathbf{H}\mathbf{x}_n$ until the rank vector converged below a given tolerance to simulate a random surfer moving through the web graph via the links. We visualized the convergence of the PageRank vector by plotting each page's rank score for each iteration, which allowed us to track how the scores changed until they reached a steady state. The resulting PageRank scores are:

$$\mathbf{x}_{\text{converged}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4/9 \\ 2/9 \\ 1/3 \end{pmatrix} \quad (3)$$

Clearly, pages 4–6 dominate the steady-state distribution, whereas Pages 1–3 decay to zero in the same limit. This behavior matches the convergence pattern shown in Figure 2.

ADD SMTH ABOUT PAGE 4 HAVING THE HIGHEST IMPORTANCE RANKING

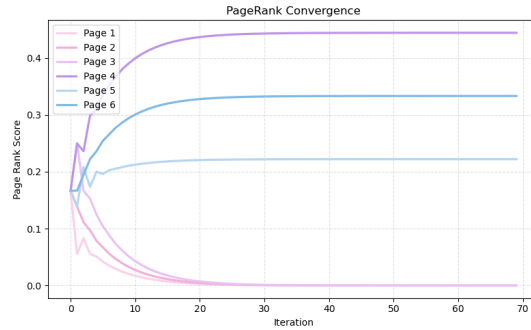


Figure 2. PageRank convergence for graph 1.

TALK ABOUT SECOND GRAPH EX HERE

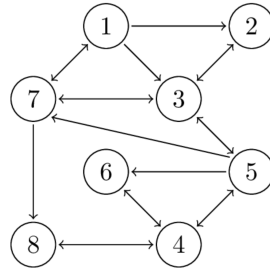


Figure 3. example graph 2.

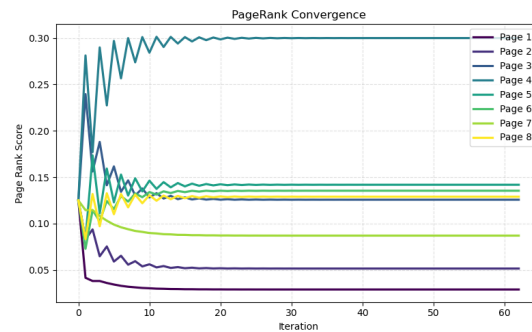


Figure 4. PageRank Convergence for graph 2.

RESULTS

results

SUMMARY AND CONCLUSION

summary/concl

Acknowledgements

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REFERENCES

- 46 Brin, S., & Page, L. 1998, Computer Networks, 30, 107.
47 <https://snap.stanford.edu/class/cs224w-readings/>
48 [Brin98Anatomy.pdf](#)
- 49 Langville, A. N., & Meyer, C. D. 2004, Internet
50 mathematics, 1, 335
- 51 —. 2009, Google’s pagerank and beyond: The science of
52 search engine rankings., 1st edn. (Princeton, N.J:
53 Princeton University Press)
- 54 Page, L., Brin, S., Motwani, R., & Winograd, T. 1999, The
55 PageRank Citation Ranking: Bringing Order to the
56 Web., Technical Report 1999-66, Stanford InfoLab.
57 <http://ilpubs.stanford.edu:8090/422/>